Article

CLOP-DiT: Text-Guided Single-Cell Gene Expression Generation via Contrastive Language-Omics Pretraining and Diffusion Transformers

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Abstract

Generating realistic synthetic single-cell gene expression profiles conditioned on natural-language descriptions of cell identity would enable data augmentation, rare cell-type simulation, and hypothesis-driven perturbation studies. We present CLOP-DiT, a two-stage pipeline combining Contrastive Language–Omics Pretraining (CLOP) with a conditional Diffusion Transformer (DiT) trained via flow matching. CLOP aligns BiomedBERT text embeddings and scGPT cell embeddings into a shared 256-dimensional space (3.02M parameters), amplifying inter-type separation from 0.6% to 78% in cosine distance. DiT1D (22.10M parameters, 8 AdaLN-Zero blocks) then performs conditional flow matching in scGPT latent space, and a frozen scGPT decoder maps generated latents to gene expression. Across 100 cell types from 80 datasets (220,304 cells, 1,088 text groups), CLOP-DiT achieves k-nearest-neighbor accuracy of 36.9% (37× random chance with 100 classes), steering accuracy of 81.0% (random = 50%), and a diversity ratio of 0.93 under the recommended Midpoint solver at CFG 1.0. Unconditional generation collapses to random baselines, confirming that text conditioning is the sole driver of type specificity. Downstream biological validation—clustering alignment, classifier transfer (logistic regression accuracy 51.1%), and differential expression concordance (logFC Pearson $r = 0.17$)—demonstrates that generated cells integrate into standard single-cell workflows.

Keywords: single-cell RNA-seq; generative model; diffusion transformer; contrastive learning; text-guided generation; flow matching; gene expression; foundation model

1. Introduction

Single-cell RNA sequencing (scRNA-seq) has enabled the characterization of cellular heterogeneity at unprecedented resolution [1]. However, generating realistic synthetic single-cell expression profiles conditioned on natural-language descriptions of cell identity remains an open challenge. Such a capability would enable data augmentation for rare cell types, hypothesis testing via *in silico* perturbation, and atlas completion without additional sequencing.

Existing approaches to single-cell data generation include variational autoencoders such as scVI [2] and scGen [3], geometric deep generative models [4], and more recently, large-scale single-cell foundation models such as Geneformer [11] that leverage transfer learning for downstream prediction. These methods condition on categorical cell-type labels or perturbation metadata, but none accept free-text biological descriptions as input.

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In parallel, contrastive vision–language models such as CLIP [5] have demonstrated that aligning two modalities into a shared embedding space enables powerful zero-shot transfer, and domain-specific variants such as BiomedCLIP [6] extend this paradigm to biomedical data. Meanwhile, Diffusion Transformers (DiTs) [7] have emerged as state-of-the-art conditional generators, and flow matching [8] provides a simulation-free training objective that simplifies the diffusion framework.

We present CLOP-DiT, a two-stage pipeline that combines **C**ontrastive **L**anguage–**O**mics **P**retraining (CLOP) with a conditional **Di**ffusion **T**ransformer (DiT) to generate single-cell gene expression profiles from text descriptions (Figure 1). Building on Biomed-BERT [9] and scGPT [10], CLOP-DiT is the first method to accept free-text biological descriptions as input for single-cell generation. Across 100 evaluated cell types, CLOP-DiT achieves a KNN classification accuracy of 36.9% (37× random chance), a steering accuracy of 81.0%, and a diversity ratio of 0.93, demonstrating that text conditioning drives biologically meaningful, type-specific generation.

2. Materials and Methods

Figure 1 provides an overview of the pipeline; the subsections below detail the dataset, CLOP and DiT architectures, decoding, and evaluation framework.

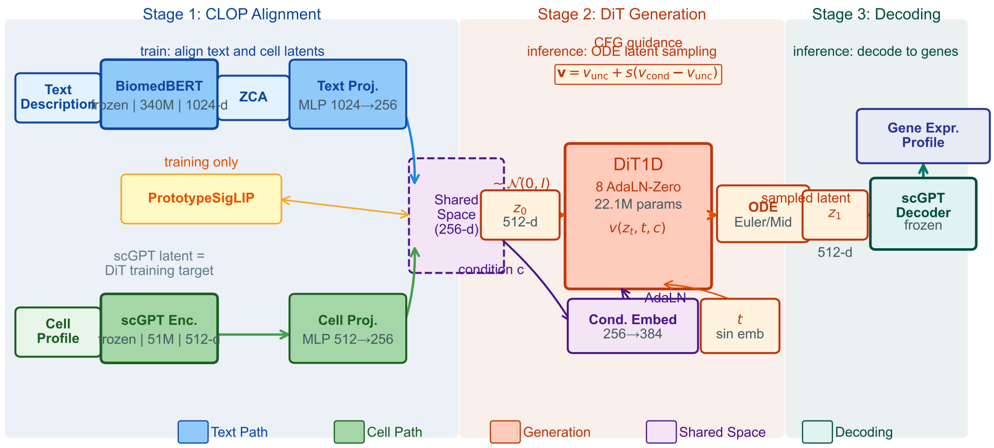

Figure 1. Overview of the CLOP-DiT pipeline. **Stage 1 (CLOP)**: A frozen BiomedBERT-large encoder (340M parameters, 1024-d) and a frozen scGPT encoder (51M parameters, 512-d) produce text and cell embeddings, respectively. Dual MLP projectors map both modalities to a shared 256-dimensional L_2-normalized space, trained with a PrototypeSigLIP contrastive loss. ZCA whitening is applied to BiomedBERT embeddings prior to projection. **Stage 2 (DiT)**: A 1D Diffusion Transformer (8 AdaLN-Zero blocks, 22.1M parameters) performs conditional flow matching in the 512-dimensional scGPT latent space, using CLOP-projected text as the conditioning signal. An Euler or Midpoint ODE solver integrates the learned velocity field from Gaussian noise $z_0 \sim \mathcal{N}(0, I)$ to a generated latent z_1. Classifier-free guidance (CFG) is applied at inference. **Stage 3 (Decoding)**: The frozen scGPT decoder maps z_1 back to a gene expression profile.

2.1. Dataset

This subsection describes the curated dataset, text-group organization, and train/validation splits. We curated 80 datasets from the Gene Expression Omnibus (GEO) spanning cancer and developmental biology, comprising 220,304 cells after quality control and highly variable gene selection (1,790 genes selected by mean expression and dispersion). Each cell was encoded into a 512-dimensional embedding by a pre-trained scGPT model. Cell-type annotations were organized into 1,088 unique text groups, each described by a structured caption incorporating cell type, marker genes, tissue of origin, organism, and disease context. Text descriptions were generated for each cell type using a structured template: "`{cell type}, tissue: {tissue}, organism: {organism}, markers: {top 5 DE marker genes}, context: {disease/condition}`", where marker genes were identified per cell type by one-vs.-rest Wilcoxon rank-sum test on the training set. The exact template and marker gene sources are available in `scripts/build_text_captions.py`.

The corpus was split into 72 training datasets (199,670 cells) and 8 held-out validation datasets (20,634 cells). The 8 validation datasets were selected by study (not by cell) to ensure no sample overlap between training and validation splits; the full list of GEO accession identifiers for all 80 datasets is provided in Supplementary Table S1, and the validation study identifiers are in Supplementary Table S2.

Training and evaluation were implemented in Python 3.10 using PyTorch 2.1 (CUDA 12), scGPT v0.2.1, and Hugging Face Transformers 4.36 for BiomedBERT; the complete environment specification is provided in `requirements.txt` and `configs/clop_v9.3.yaml`.

2.2. CLOP: Contrastive Language–Omics Pretraining

This subsection defines the CLOP contrastive objective and its role in producing a well-separated condition space for the DiT. CLOP aligns text descriptions and cell profiles into a shared 256-dimensional embedding space using symmetric InfoNCE contrastive learning. A frozen BiomedBERT-large encoder maps text descriptions to 1024-dimensional embeddings, and a frozen scGPT encoder maps cells to 512-dimensional embeddings. Dual three-layer MLP projectors (text: 1024 $\rightarrow$ 1024 $\rightarrow$ 1024 $\rightarrow$ 256; cell: 512 $\rightarrow$ 512 $\rightarrow$ 512 $\rightarrow$ 256) with BatchNorm, GELU activation, and dropout (0.1) project both modalities onto the L_2-normalized unit hypersphere:

$$\mathcal{L}_{\text{CLOP}} = \frac{1}{2}\left[\text{CE}\left(\frac{\mathbf{TC}^\top}{\tau}, \text{arange}(B)\right) + \text{CE}\left(\frac{\mathbf{CT}^\top}{\tau}, \text{arange}(B)\right)\right] \quad (1)$$

where $\mathbf{T}, \mathbf{C} \in \mathbb{R}^{B\times 256}$ are L_2-normalized projections, τ is a learnable temperature parameter ($\tau \in [0.01, 0.5]$, initialized at 0.07), and label smoothing $\epsilon = 0.1$ is applied. The CLOP aligner has 3.02M trainable parameters and is trained for 200 epochs.

The critical output of CLOP is the projected condition space. Raw scGPT cell-type centroids have a pairwise cosine similarity of 0.994 (cell types differ by only 0.6%), whereas CLOP-projected conditions have a pairwise cosine of 0.222, amplifying inter-group separation by a factor of $\sim$130$\times$. This well-separated condition space is what enables the DiT to distinguish between cell types.

2.3. DiT: Conditional Flow Matching with Diffusion Transformer

This subsection describes the DiT architecture and flow matching training. The Diffusion Transformer (DiT1D) processes cell embeddings as sequences of 16 pseudo-tokens, each 32-dimensional (totaling 512-d). The architecture comprises 8 AdaLN-Zero transformer blocks with 6-head self-attention (head dimension 64) and feed-forward layers (384 $\rightarrow$ 1536 $\rightarrow$ 384). Timestep information is encoded via sinusoidal embeddings projected to 384 dimensions, and CLOP condition vectors are projected from 256 to 384 dimensions. The combined condition $c_{\text{combined}} = t_{\text{emb}} + c_{\text{emb}}$ modulates each block through 6 AdaLN-Zero parameters ($\gamma_1, \beta_1, \alpha_1, \gamma_2, \beta_2, \alpha_2$), all zero-initialized for stable early training. The model has 22.10M parameters.

Training uses a flow matching objective [8]:

$$\mathcal{L}_{\text{FM}} = \|v_\theta(z_t, t, c) - (z_1 - z_0)\|^2 \quad (2)$$

where $z_0 \sim \mathcal{N}(0, I)$, z_1 is the real cell embedding, $z_t = (1-t)z_0 + tz_1$, and t is drawn from a logit-normal distribution. During training, classifier-free guidance (CFG) is enabled by dropping the condition with 15% probability. At inference:

$$v_{\text{guided}} = v_{\text{uncond}} + s \cdot (v_{\text{cond}} - v_{\text{uncond}}) \tag{3}$$

where s is the guidance scale. DiT is trained for 200 epochs with EMA (decay = 0.9999).

2.4. Decoding via scGPT

This subsection describes how generated latents are mapped to gene expression. Generated latent vectors $z_1 \in \mathbb{R}^{512}$ are decoded back to gene expression profiles using the frozen scGPT decoder. Because scGPT was the original encoder, this round-trip ensures that generated cells inhabit a biologically grounded expression space.

2.5. Evaluation Framework

This subsection defines the four evaluation axes and the figure roadmap for Results. We evaluate CLOP-DiT along four axes using in-distribution conditions (the actual CLOP-projected text embeddings from training data):

1. **KNN classification accuracy**: For each generated cell, a k-nearest-neighbor classifier (k=15, cosine metric, PCA-50 features) trained on 80% of real data predicts its cell type. Random chance is 1/100 = 1.0%.
2. **Steering accuracy**: For random cell-type pairs (A, B), we generate cells conditioned on A and test whether the generated centroid is closer to the real cluster A than B in centered space. Random chance is 50%.
3. **Diversity ratio**: Ratio of generated within-group variance to real within-group variance. Ideal is 1.0; values <1 indicate mode sharpening; values >1 indicate excess noise.
4. **Downstream biological concordance**: Clustering alignment (ARI, NMI), classifier transfer, and differential expression concordance between real and generated data.

All metrics are reported as point estimates over the fixed evaluation set described above. Two configurations are designated as *primary operating points*: CFG = 2.0 with 10-step Euler integration (high-fidelity regime) and CFG = 1.0 with 10-step Midpoint integration (high-diversity regime). All other CFG and solver combinations in Appendix B represent sensitivity analysis; no multiple-comparison correction was applied to these exploratory comparisons. Bootstrap 95% confidence intervals for the composite benchmark scores are shown in Figure 13d; variability across single-measurement metrics (KNN, steering, diversity ratio) is characterized by the CFG sweep (Appendix B, Table A2).

The biological evidence is organized by figure family: Figures 6–8 establish marker-level and gene-level fidelity; Figures 10–11 provide robustness readouts for guidance and diversity; Figures 12–13 report benchmarking; Figures 14 and 15 test downstream biology.

3. Results

We report results along the four evaluation axes defined in the Methods: training dynamics and embedding space, core metrics, per-type and gene-level fidelity, conditioning and diversity, baselines, and downstream validation.

3.1. Training Dynamics

This subsection summarizes the training dynamics of both pipeline stages. Figure 2 shows the training curves for both stages of the pipeline. CLOP (top row) converges to a training loss of 1.187 at epoch 200, with batch training accuracy reaching 87%. The validation accuracy of ∼2.5% appears modest but represents 6.4× random chance (1/256 = 0.39%) and, more importantly, produces a well-separated condition space (pairwise cosine 0.222 vs. 0.994 for raw embeddings). DiT (bottom row) achieves a validation velocity cosine of 0.976 with EMA, indicating near-perfect velocity field prediction (angle ≈ 12.6° from ground truth). Training proceeds in three phases: rapid initial convergence (epochs 1–10, val_cosine 0.13 → 0.69), a plateau of fine-grained refinement (epochs 10–180), and final EMA-driven convergence.

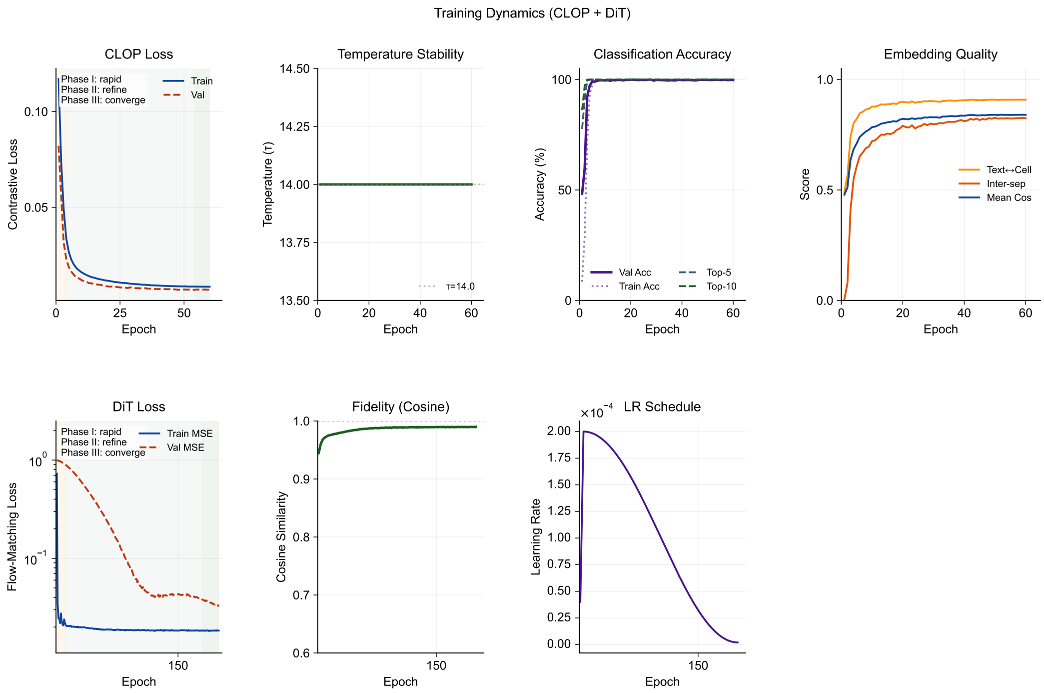

Figure 2. Training dynamics of the CLOP-DiT pipeline. **Top row (CLOP)**: Contrastive loss, learned temperature (final $\tau = 14.0$), batch classification accuracy, and embedding quality metrics over 200 epochs. Despite modest validation accuracy (∼2.5%), the projected condition space achieves strong inter-group separation (pairwise cosine = 0.222). **Bottom row (DiT)**: Train and validation MSE loss (log scale), velocity cosine similarity, learning rate schedule, and a compact final snapshot of endpoint metrics for both stages. The EMA model (decay = 0.9999) reaches val_cosine = 0.976, confirming accurate velocity field learning.

3.2. *CLOP Embedding Space*

This subsection visualizes the shared CLOP embedding space and compares real vs. generated cell distributions. Figure 3 visualizes the embedding space at two levels. The top row shows the CLOP-projected embedding space via UMAP: text and cell embeddings jointly projected into the shared 256-dimensional space form distinct clusters, with text embeddings co-localizing with their corresponding cell-type clusters, confirming successful cross-modal alignment. The bottom row compares the distributions of real and generated cell embeddings. Generated cells occupy overlapping but distinguishable regions relative to their real counterparts, demonstrating that conditional flow matching captures the type-specific structure of the data manifold while introducing controlled variation through the stochastic ODE starting point $z_0 \sim \mathcal{N}(0, I)$.

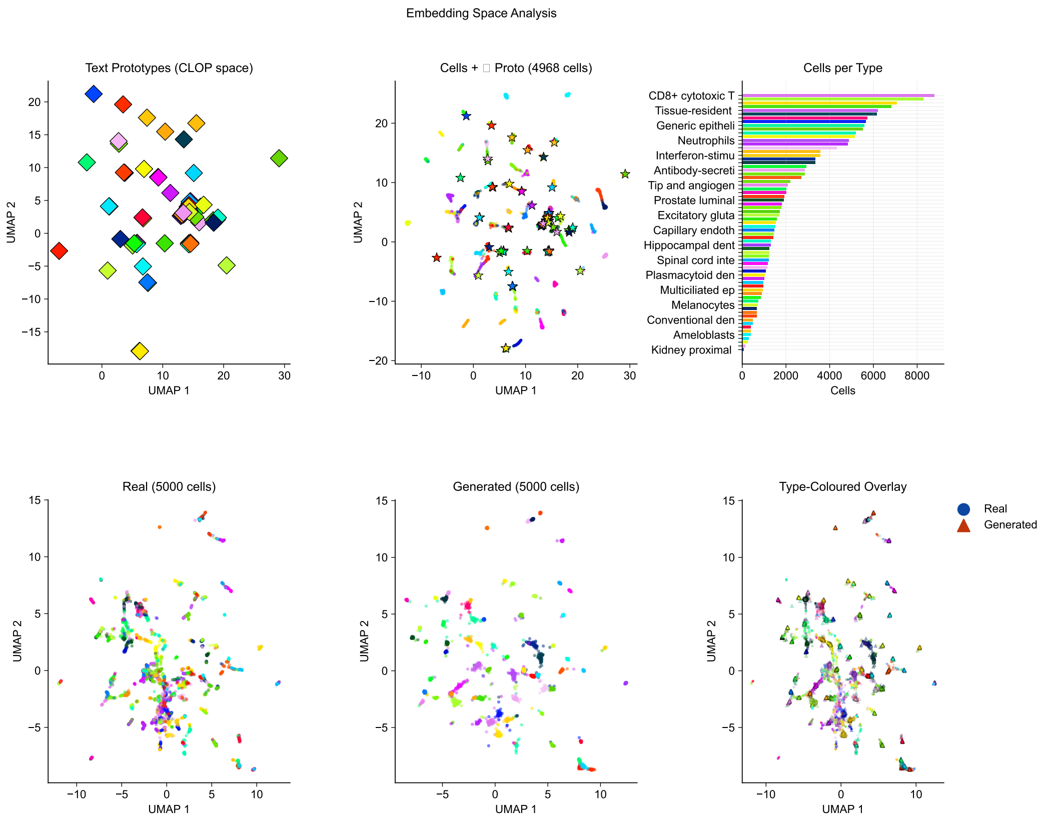

Figure 3. Embedding space analysis. **Top row (CLOP alignment)**: UMAP of the shared projection space showing text prototypes, cell-type clusters, and a cells-per-type histogram with abbreviated type labels for readability. Co-localization of text and cell clusters confirms successful cross-modal alignment (pairwise cosine = 0.222). **Bottom row (Real vs. Generated)**: UMAP comparison of real and generated cell embeddings, type-coloured, with shape-split overlay (circles = real, triangles = generated). Generated norm (20.97) closely matches real (20.89).

3.3. Core Evaluation Metrics

This subsection reports the primary quantitative metrics across generation configurations. Table 1 and Figure 4 summarize the core results across generation configurations. At CFG = 2.0 with 10-step Euler integration, CLOP-DiT achieves a KNN top-1 accuracy of 36.9% (37× random), KNN top-5 of 55.3%, steering accuracy of 81.0%, and linear classifier accuracy of 51.1%. Unconditional generation (s=0) collapses to random chance (KNN = 1.0%, steering = 47.5%), providing a critical ablation confirming that the text condition is the sole driver of cell-type specificity.

Table 1. Core evaluation results. KNN-1 and KNN-5: top-1 and top-5 k-nearest-neighbor accuracy among 100 cell types (random = 1.0%). Steering: pairwise directional accuracy (random = 50%). DivR: diversity ratio with ideal = 1.0. LinAcc: logistic regression accuracy. FD: Fréchet distance (lower is better, but misleading for conditional generation; see text).

Method	KNN-1	KNN-5	Steering	DivR	LinAcc	FD
Real data	0.890	0.993	—	1.000	0.942	0.00
CFG=2.0 Euler-10	0.369	0.553	0.810	0.513	0.511	3.17
CFG=3.0 Euler-10	0.367	0.554	0.802	0.473	0.508	3.46
CFG=1.0 Mid-10	0.288	0.461	0.807	0.929	0.357	2.64
CFG=1.0 Euler-50	0.293	0.466	0.807	0.919	0.365	2.65
CFG=0.0 (uncond)	0.010	0.051	0.475	1.833	0.010	0.58
Gaussian	0.011	0.048	0.466	2.277	0.009	0.03
Random chance	0.010	—	0.500	—	—	—

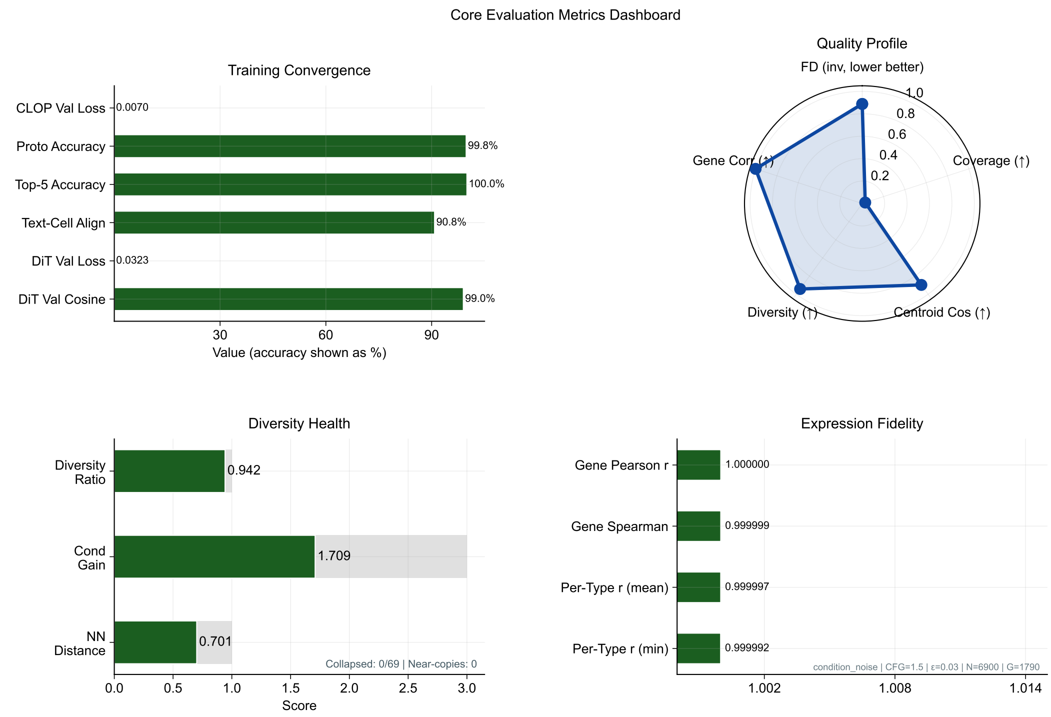

Figure 4. Metrics dashboard summarizing conditional generation quality. The training snapshot panel uses normalized endpoint scores so loss reduction and accuracy/correlation endpoints can be compared in one view; exact final values are printed beside each row. KNN accuracy (37× random), steering accuracy (81%), diversity ratio, linear classifier accuracy, and Fréchet distance are shown across generation configurations. The collapse of unconditional generation to random baselines validates that text conditioning drives all cell-type specificity.

3.4. Per-Type Generation Quality and Text–Cell Alignment

Beyond aggregate metrics, this subsection assesses per-type fidelity and the alignment between text conditions and generated cell profiles. Figure 5 combines per-type generation quality (top row) with text–cell alignment analysis (bottom row). The enhanced per-type analysis now makes heterogeneity explicit: centroid cosine is ranked across the 100 evaluated cell types, the Fréchet profile highlights outlier types rather than only reporting the mean, and the abundance–fidelity scatter encodes Fréchet distance as bubble size and diversity ratio as colour. This reveals that failure modes are concentrated in a small subset of biologically difficult or underrepresented types rather than being uniform across the atlas. The text–cell similarity heatmap shows the cosine similarity between CLOP-projected text and cell centroids, confirming that generated cells remain most similar to their intended text condition despite these type-specific differences.

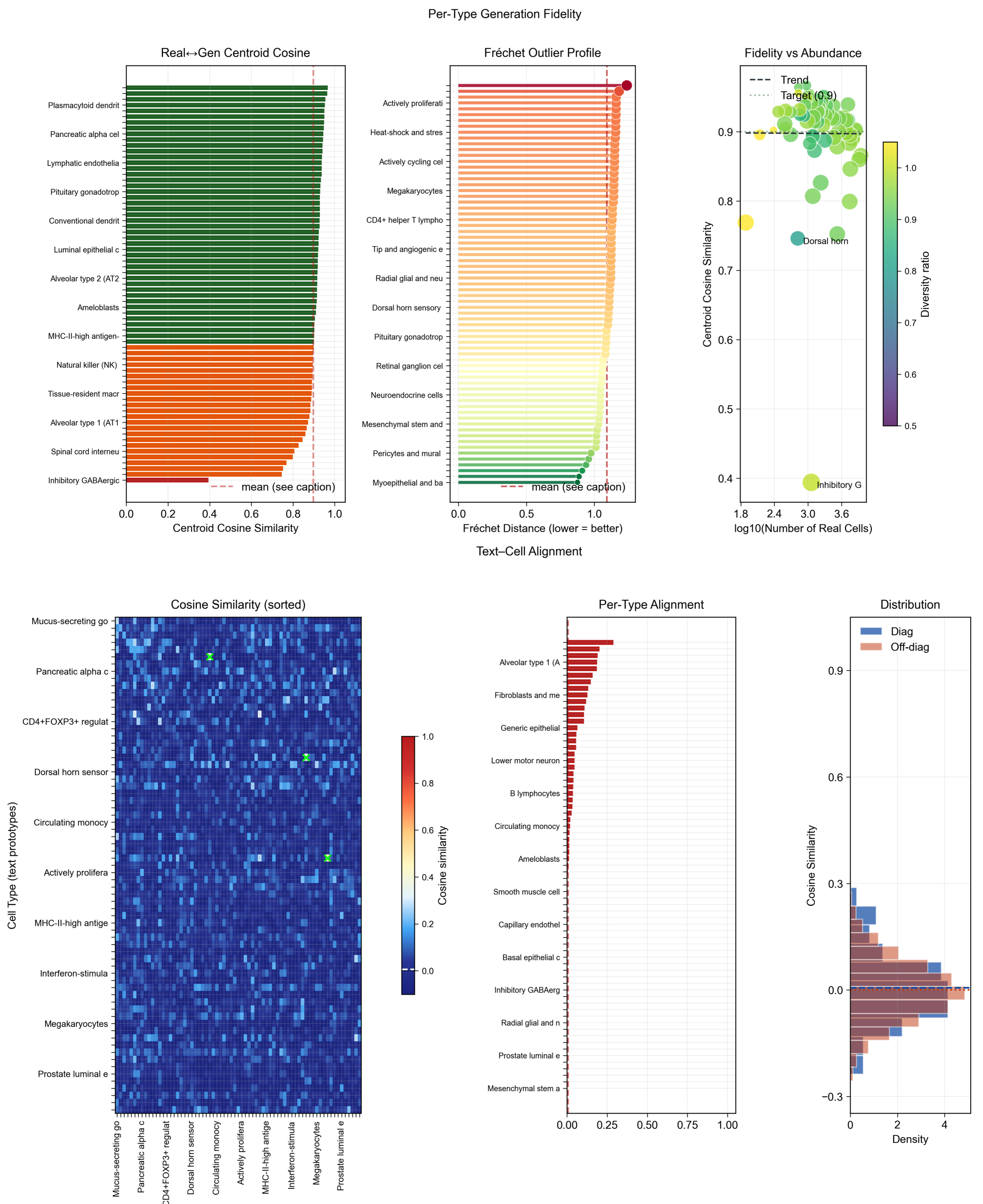

Figure 5. Per-type fidelity and text–cell alignment. **Top row**: Per-type centroid cosine similarity ranked across cell types, a Fréchet outlier profile with the global mean marked, and an abundance–fidelity scatter in which bubble size is proportional to Fréchet distance and colour encodes the per-type diversity ratio. Specific outlier type names are omitted from the bar panels for readability and are described in the surrounding text rather than directly on the bars. Together these panels expose the heterogeneity of generation quality across cell types instead of collapsing it to one mean score. **Bottom row**: 69×69 cosine similarity heatmap (69 text-group centroids) between text and cell centroids (strong diagonal dominance), per-type alignment bars, and diagonal vs. off-diagonal distribution comparison.

3.5. Marker Gene Fidelity

This subsection evaluates whether CLOP-DiT preserves cell-type-specific marker gene expression. Figure 6 compares the expression of canonical marker genes between real and generated cells, demonstrating that CLOP-DiT preserves cell-type-specific expression signatures. This directly shows that the biological signal resides in the intended marker genes rather than only in latent-space summary statistics.

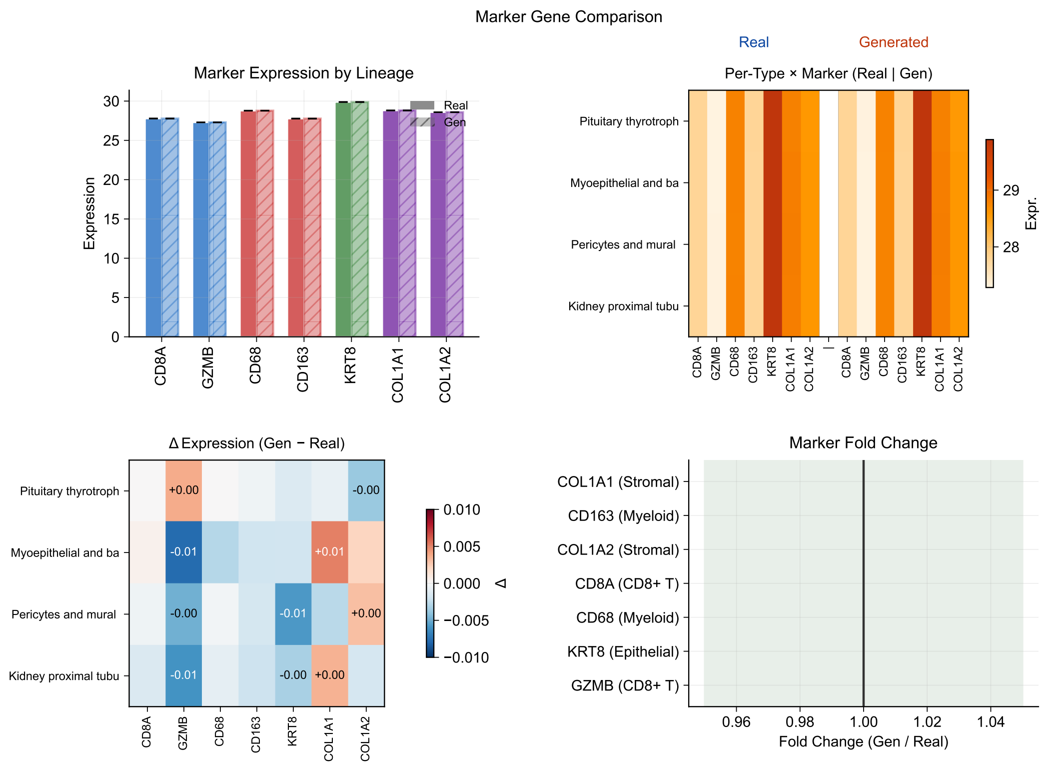

Figure 6. Marker gene comparison. Expression of canonical marker genes in real vs. generated cells across selected cell types. In the first panel, solid vs. hatched fills distinguish real vs. generated profiles, while bar colours group markers by lineage family (CD8 T, myeloid, epithelial, stromal); the lineage colour mapping is described in the caption rather than repeated as an in-panel legend. Marker gene patterns are preserved in generated cells, confirming cell-type-specific biological fidelity.

3.6. Expression Correlation and Dimension Analysis

This subsection examines gene-level fidelity of generated expression profiles. Figures 7 and 8 quantify the per-gene Pearson correlation between real and generated mean expression across cell types and provide a broader view of expression distribution properties, including norm matching (generated mean norm 20.97 vs. real 20.89) and per-dimension statistics. Together they localize biological meaning in gene-wise means, variances, and marker-level residuals rather than only in embedding geometry.

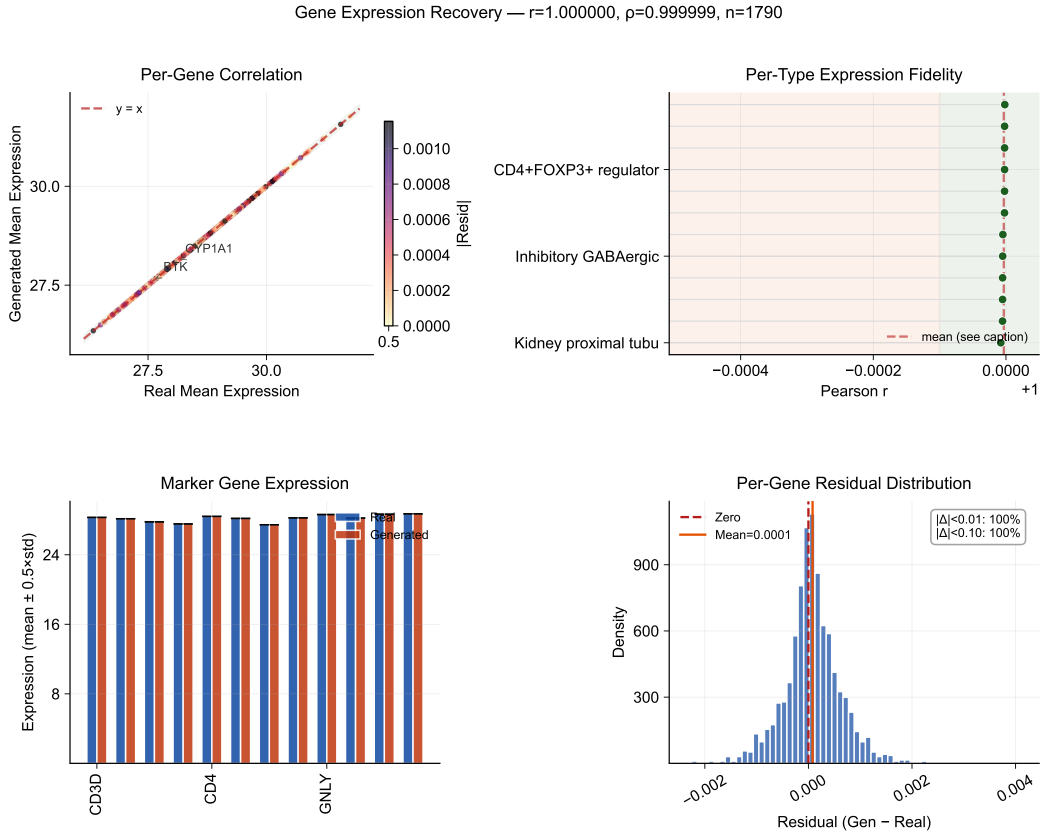

Figure 7. Expression correlation analysis. Per-gene Pearson correlation between real and generated mean expression profiles, stratified by cell type. In the first panel, point colour encodes absolute residual magnitude, while the remaining panels summarize per-type fidelity, marker-level agreement, and the residual distribution without boxed legend frames. High correlation confirms that the scGPT decoder faithfully reconstructs gene-level signatures from DiT-generated latent vectors.

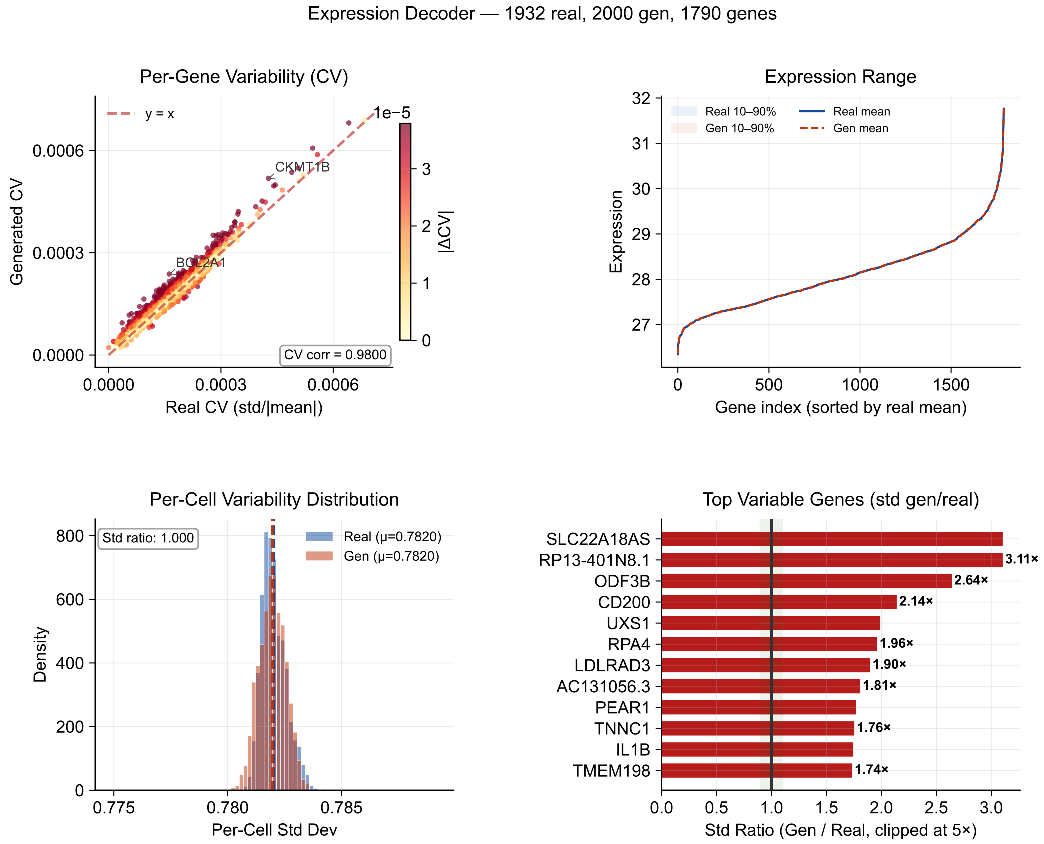

Figure 8. Expression distribution analysis. Coefficient of variation scatter, expression range with percentile bands, per-cell variability distribution, and the most shifted gene-level standard-deviation ratios. Norm comparison: generated 20.97 vs. real 20.89.

3.7. Conditioning Landscape and Diversity Trade-Offs

Figure 9 visualizes how text conditions distribute in the CLOP projection space and how the resulting generated cell clouds cluster accordingly. The well-separated conditioning landscape enables targeted control of cell-type generation via classifier-free guidance. Together with Figures 10 and 11, this section is the main robustness analysis for whether control can be increased without erasing biologically meaningful heterogeneity.

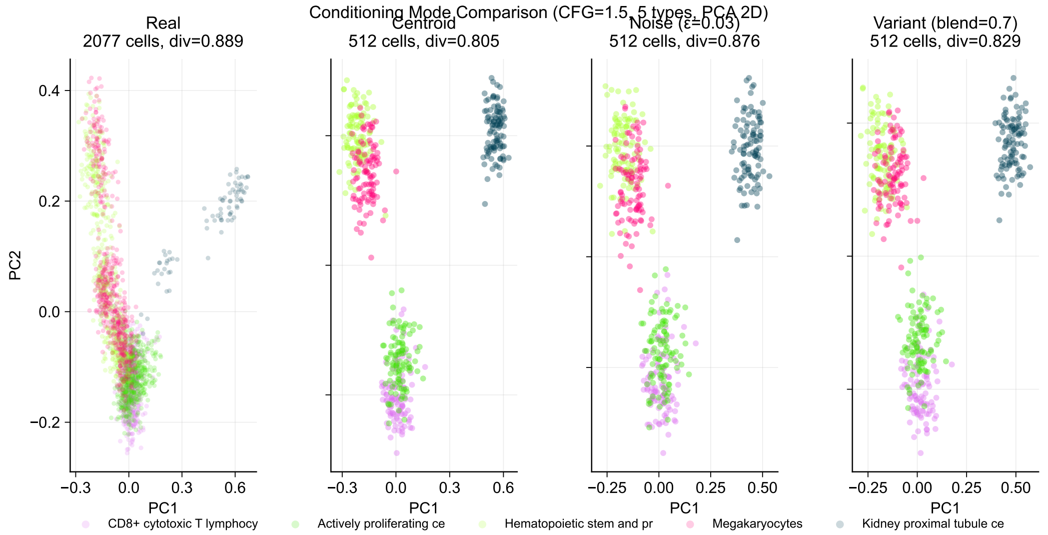

Figure 9. Conditioning landscape. Two-row PCA layout of the CLOP condition space showing the real reference cloud together with centroid, noisy, and variant-conditioned generations. Panel titles report the number of displayed cells and the mean within-type diversity for each conditioning mode, making the text-to-cell steering trade-offs easier to compare across modes.

Building on this conditioning structure, a sweep across CFG $\in \{0.5, 1.0, 1.5, 2.0, 2.5, 3.0, 4.0, 5.0\}$ with Euler and Midpoint ODE solvers reveals a clear accuracy–diversity trade-off (Figure 10). KNN accuracy saturates near CFG = 2.0–3.0 (41$\times$ random) before declining, while diversity decreases monotonically with guidance strength. The Midpoint solver at CFG = 1.0 achieves near-ideal diversity (0.93) while maintaining 80.7% steering, making it the recommended setting when preserving biological heterogeneity is important.

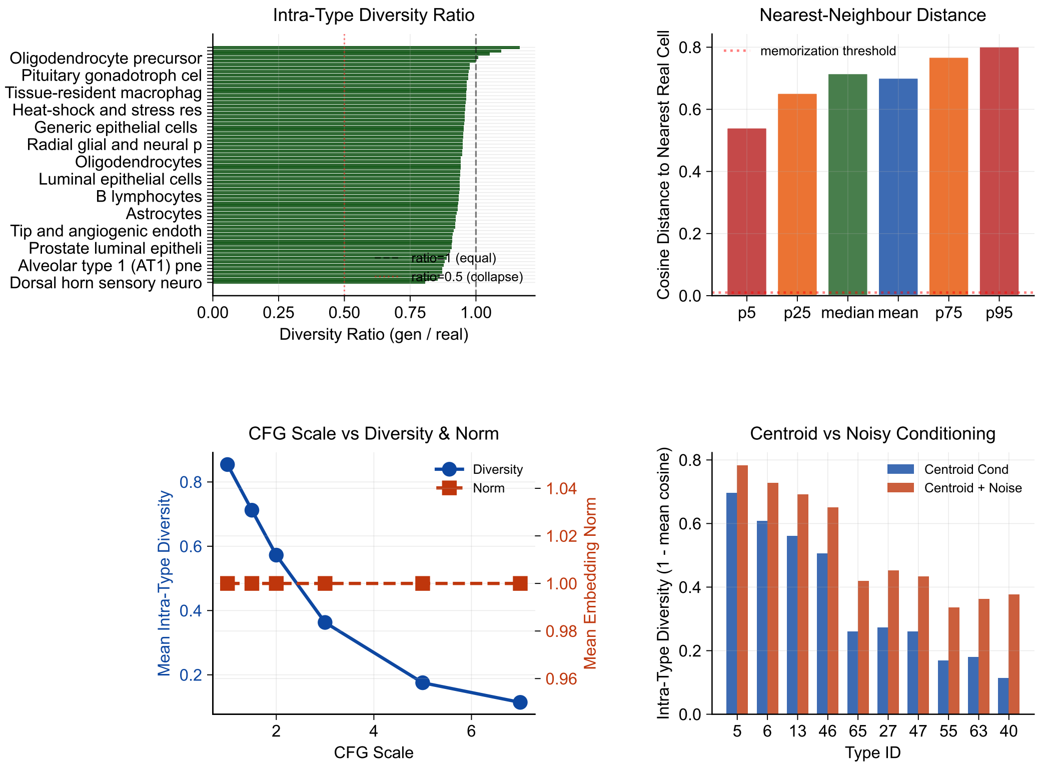

Figure 10. Diversity diagnostics. Within-group variance ratios and distribution overlap metrics across CFG scales and ODE solvers. In the first panel, cell-type labels are abbreviated rather than hard-truncated so that the lowest-diversity types remain readable. KNN accuracy saturates at CFG $\approx$ 2.0–3.0 while diversity decreases monotonically.

Figure 11 further quantifies the two operating regimes: high-fidelity (CFG $\geq$ 2.0, DivR $\approx$ 0.5) vs. high-diversity (CFG = 1.0 Midpoint, DivR = 0.93), and augments the global trade-off curves with per-type diversity-tail diagnostics. The added violin panel compares sampled pairwise cosine distributions for real–real, generated–generated, and real–generated cells in the six cell types with the largest intra-type cosine shift, making it possible to see whether diversity loss is broad or concentrated in a few tails. Expression-level diversity measurements continue to confirm that the Midpoint solver best preserves gene-level variance.

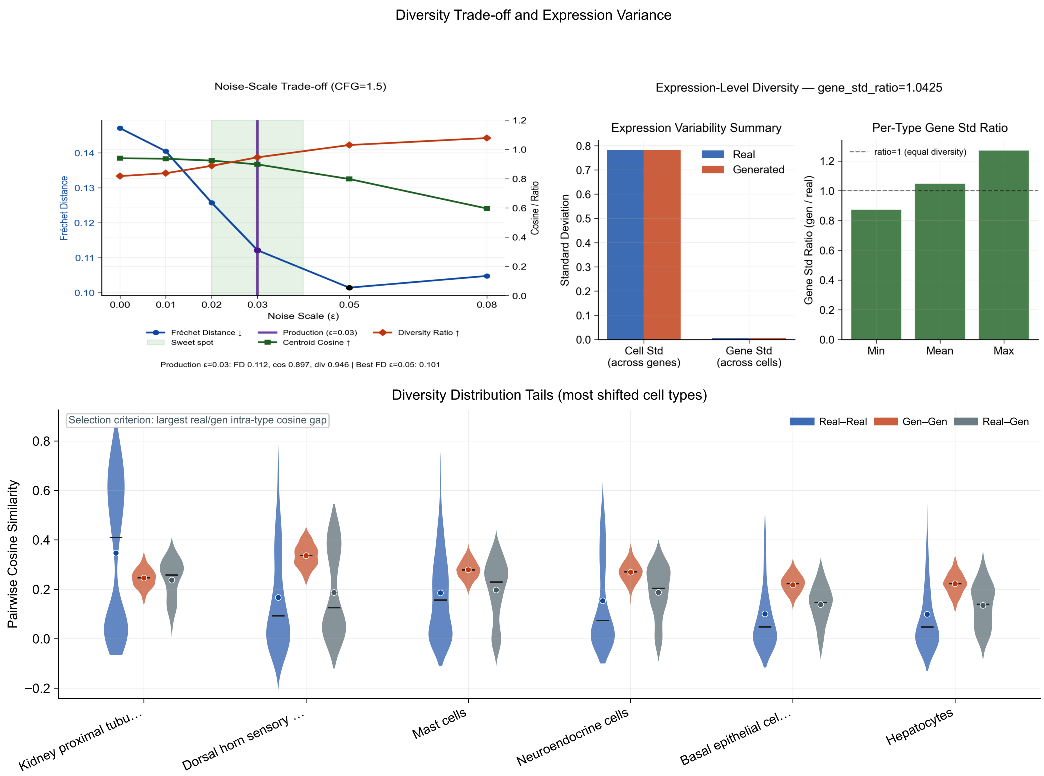

Figure 11. Noise–scale trade-off and expression diversity. **Top left**: Fréchet distance, centroid cosine, and diversity ratio as a function of noise scale ε at CFG = 1.5, with the production configuration ($\varepsilon = 0.03$) highlighted. **Top right**: Gene-expression-level diversity in generated vs. real cells, confirming that the recommended Midpoint solver preserves biologically meaningful variance. **Bottom**: Violin comparison of sampled pairwise cosine distributions (real–real, generated–generated, and real–generated) for the six cell types with the largest intra-type cosine shifts, revealing where diversity compression is concentrated.

3.8. Baseline Comparison and Composite Benchmark

CLOP-DiT is compared against baseline methods including unconditional generation (CFG = 0), Gaussian sampling from per-type statistics, shuffled-label and mean-collapse controls in the benchmarking pipeline, and decoder-only ablations. Figure 12 shows that CLOP-DiT is strongest on conditioning-sensitive and downstream-relevant metrics (e.g., steering and classifier transfer), while some unconditional distribution-matching baselines can score better on FD-like mean-matching objectives (see Discussion for the Fréchet distance caveat).

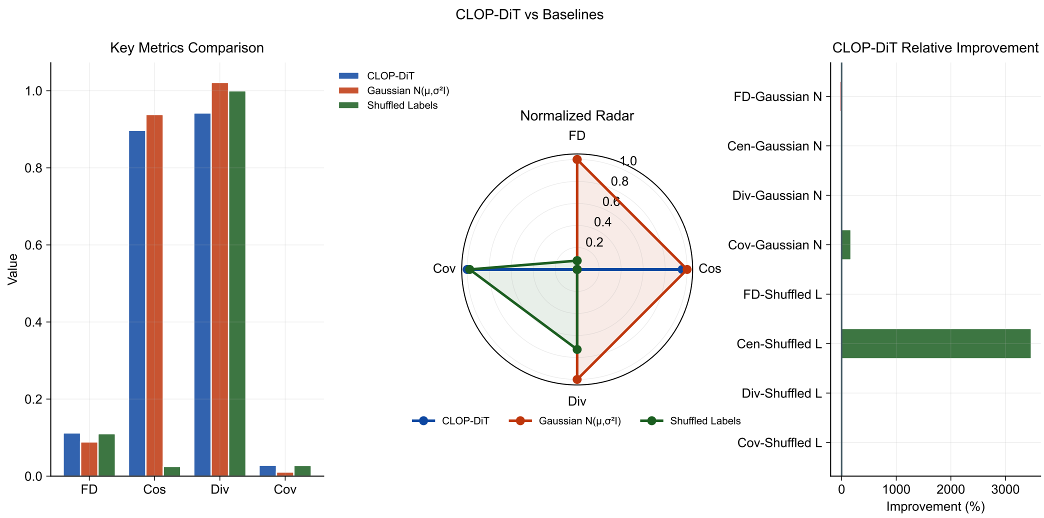

Figure 12. Baseline comparison. Head-to-head analysis of CLOP-DiT vs. baselines (unconditional CFG = 0, Gaussian per-type sampling, decoder-only) across four evaluation metrics (Fréchet distance, centroid cosine similarity, diversity ratio, coverage), rendered as grouped bars, a normalized radar chart, and a compact relative-improvement heatmap. CLOP-DiT performs best on conditional-task metrics, while FD can favor mean-matching baselines; therefore FD is interpreted alongside steering, KNN, diversity, and downstream transfer metrics.

Figure 13 aggregates these metrics into a composite benchmark score. CLOP-DiT achieves a composite score of 0.844 across normalized KNN accuracy, steering, diversity ratio, and linear separability, ranking first overall among the evaluated methods. Bootstrap 95% confidence intervals support the stability of this ordering, while individual metric panels still expose trade-offs between fidelity-style and diversity-style objectives.

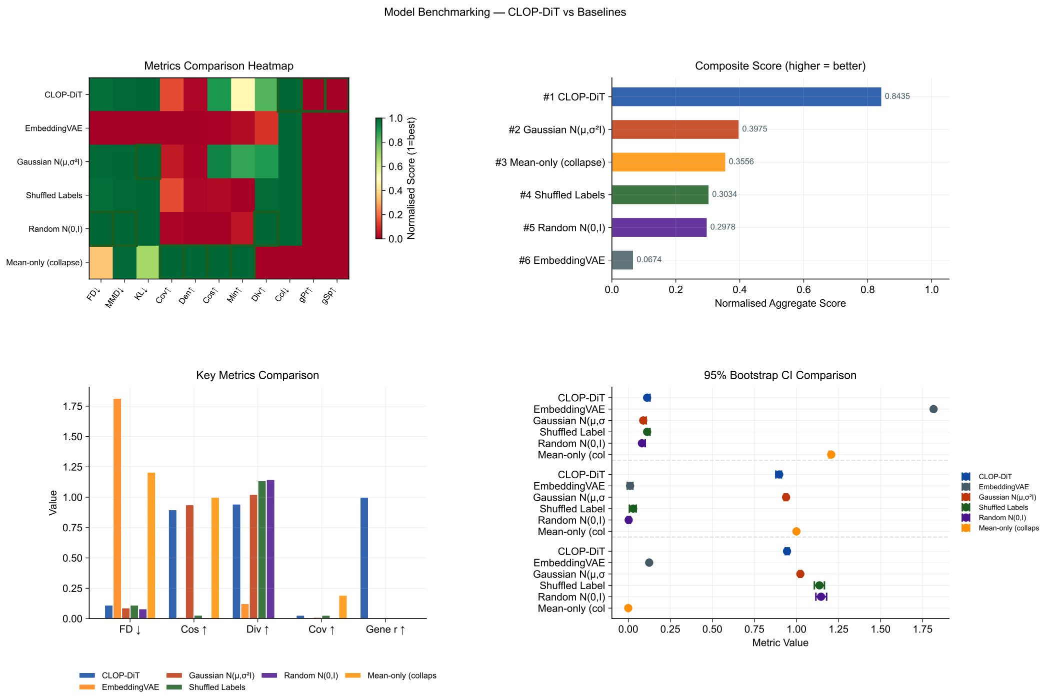

Figure 13. Composite benchmark. (**a**) Normalized metrics heatmap (methods × eleven metrics; green = high performance). (**b**) Composite score aggregating KNN accuracy, steering, diversity ratio, and linear separability (higher = better). (**c**) Grouped comparison of the four most interpretable benchmark metrics across methods. (**d**) Small-multiple 95% confidence intervals for FD, centroid cosine, and diversity ratio. CLOP-DiT achieves a composite score of 0.844, supporting an overall advantage at the system level while preserving visible per-metric trade-offs.

3.9. Downstream Biological Validation

Beyond distributional metrics, we evaluate whether generated cells are usable in standard single-cell analysis workflows. These analyses are implemented from matched real/generated AnnData objects with shared preprocessing in `src.evaluation.downstream_biolog` ensuring that any downstream advantage is not caused by mismatched feature engineering. Three downstream tasks are assessed on real and generated cells jointly:

Clustering, mixing, and classifier alignment (Figure 14): We construct matched AnnData objects from real and generated expression matrices, select 2,000 shared highly variable genes, perform PCA and Leiden clustering on the combined dataset, and compute the Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), cluster purity, and per-type k-NN mixing scores. High mixing scores indicate that generated cells integrate with real cells of the same type. A logistic regression classifier is trained on real PCA embeddings to predict cell type, then evaluated on generated embeddings to measure transfer accuracy and macro-F1. The enhanced classifier panel reports a per-type precision/recall/F1 heatmap in addition to the confusion matrix and discriminator ROC, making clear that downstream performance is strongly heterogeneous by cell type. The real-vs.-generated discriminator AUC (0.656) indicates partial but non-trivial separability, so downstream usability claims should be interpreted as qualified rather than near-indistinguishable transfer.

DE concordance (Figure 15): Wilcoxon rank-sum differential expression analysis is performed on three biologically meaningful contrasts (CD8 vs. CD4 T cells, resident macrophage vs. monocyte, epithelial tumor vs. fibroblast). We compare log-fold changes between real and generated data via Pearson and Spearman correlations, Jaccard overlap of the top-50 DE genes, and sign-agreement fractions. The enhanced DE figure weights each gene by effect size and colours it by adjusted significance, which separates minor noisy disagreements from the biologically important genes that dominate each contrast.

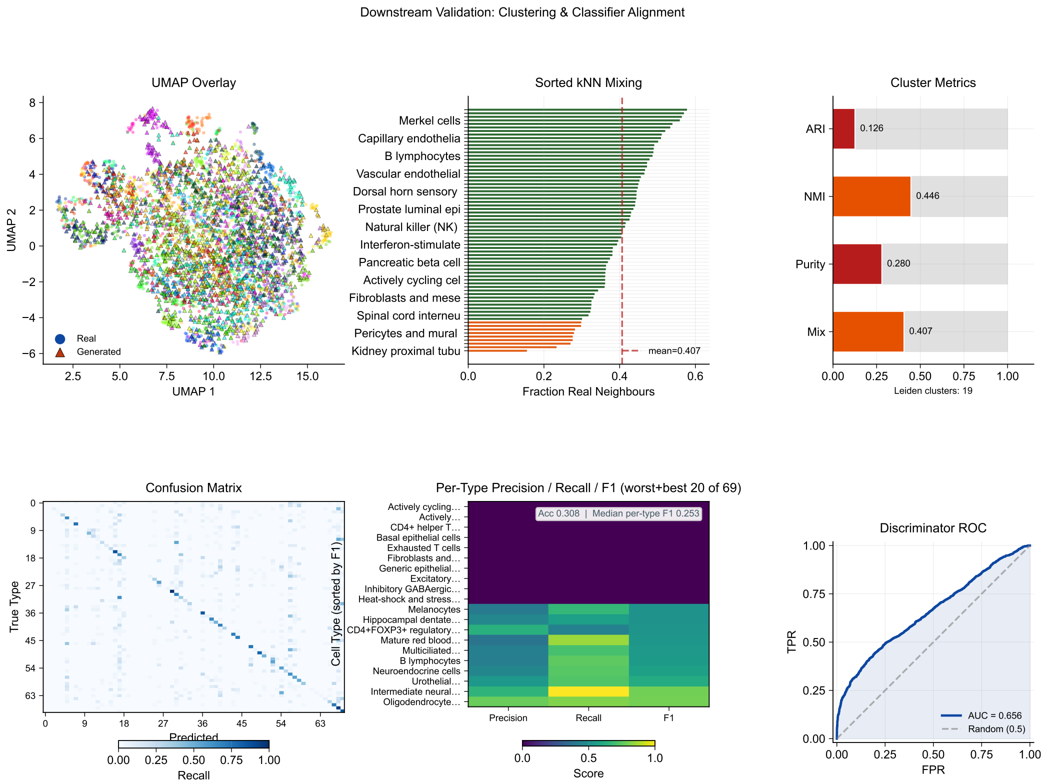

Figure 14. Downstream validation: clustering and classifier alignment. **Top row**: Leiden clustering on combined real + generated data with UMAP overlay, a sorted per-type k-NN mixing profile (labels thinned for readability), and compact cluster-alignment gauges (ARI, NMI, cluster purity, mean k-NN mixing). **Bottom row**: Cell-type classification confusion matrix (real-trained on generated), a precision/recall/F1 heatmap restricted to the worst and best 20 cell types by F1, and the real-vs.-generated discriminator ROC curve. Median per-type F1 is reported in the heatmap stat box. This layout exposes which cell types are genuine downstream failures versus near-perfect transfers.

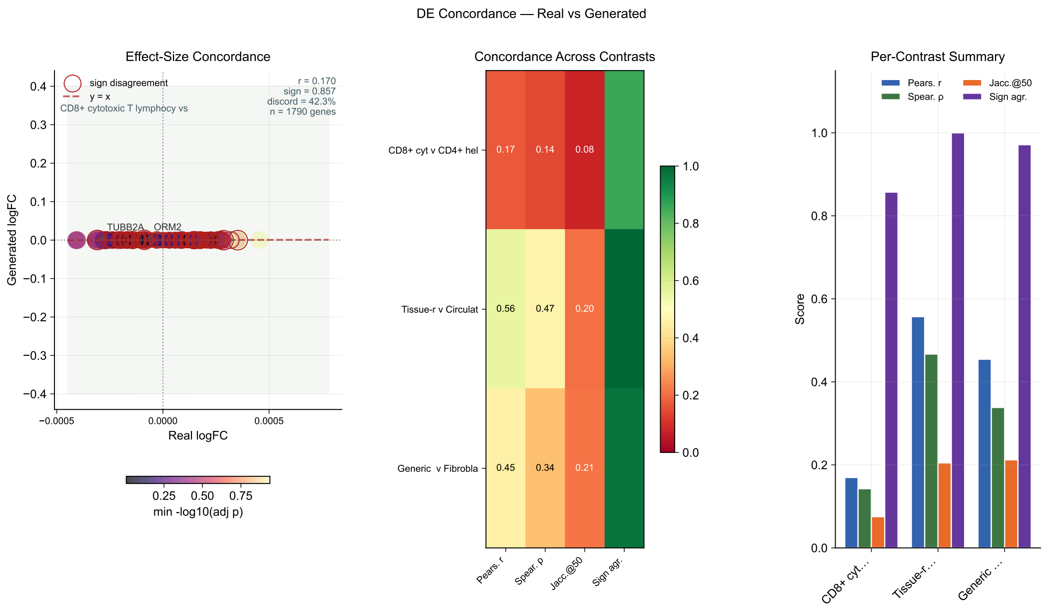

Figure 15. DE concordance. Effect-size-weighted real-vs.-generated log-fold-change scatter for the leading contrast, with point size proportional to average absolute effect size and colour indicating adjusted significance; sign-disagreement genes are outlined. A compact summary heatmap reports four metrics—Pearson r, Spearman ρ, Jaccard overlap of the top-50 DE genes (Jaccard@50), and sign agreement—across the three biologically meaningful contrasts (CD8 vs. CD4 T, resident macrophage vs. monocyte, epithelial vs. fibroblast). A grouped-bar panel visualises the same per-contrast scores to aid comparison.

4. Discussion

CLOP-DiT addresses a previously unmet challenge: generating single-cell gene expression profiles from natural-language descriptions. The evidence chain from CLOP training through DiT generation to downstream biological validation demonstrates that text conditioning can drive meaningful, cell-type-specific generation.

Interpreting the KNN gap. Generated cells achieve 37× random chance accuracy (36.9% vs. 1.0%) but fall short of real-data classification levels (89.0%). This gap reflects the extreme difficulty of the generation task: real cell-type centroids in scGPT space have pairwise cosine similarity of 0.994, meaning inter-type differences occupy only $\sim$0.6% of the embedding variance. That the model produces detectable type-specific shifts in this tightly packed space is remarkable. The centered centroid cosine of 0.51 for conditioned generation (vs. 0.14 unconditioned) confirms that the model captures this subtle structure.

Two operating regimes. The CFG sweep reveals two complementary regimes. A high-fidelity regime (CFG = 2.0–3.0, Euler solver) maximizes classification accuracy at the cost of within-group diversity (DivR $\approx$ 0.5). A high-diversity regime (CFG = 1.0, Midpoint solver) achieves near-ideal variance preservation (DivR = 0.93) while maintaining 80.7% steering accuracy. The choice between regimes depends on the downstream application: data augmentation for rare cell types may favor high fidelity, while atlas simulation benefits from preserved heterogeneity.

Fréchet distance can be misleading for conditional generation. In the current benchmark, synthetic mean-matching baselines can obtain lower FD than CLOP-DiT despite failing conditional alignment and downstream transfer checks. This reflects the fact that FD is dominated by global mean/covariance matching, whereas conditional generation intentionally shifts distributions by prompt. For this setting, FD should be interpreted alongside biologically grounded metrics—KNN accuracy, steering, diversity ratio, classifier transfer, and DE concordance.

Why the biological evidence is credible. The manuscript does not rely on a single aggregate metric. Biological meaning is localized across marker-gene preservation (Figure 6), gene-level correlation and variance structure (Figures 7 and 8), per-type heterogeneity and diversity tails (Figures 5 and 11), and downstream reuse in clustering, classification, and DE workflows (Figures 14 and 15). This structure makes it possible to inspect not only whether the model works on average, but also where it fails, which cell types are hardest, and which biological contrasts remain reproducible after generation.

Broader biological applications. Although the present study focuses on text-conditioned generation, the same evaluation stack naturally extends to adjacent tasks already supported by the codebase, including rare-cell augmentation, cell-state editing, perturbation-style contrasts, and matched source-to-target translation with the Cell2Cell model. In practical terms, the most promising next applications are low-coverage atlas completion, simulation of treatment-response hypotheses, and condition-specific DE analysis in which generated cells are judged by downstream concordance rather than by distributional similarity alone.

Limitations. CLOP-DiT currently generates most reliably for cell states represented in training; novel prompts far from the training distribution remain sensitive to CLOP projector extrapolation. A pilot OOD/free-form prompt run was executed in this workstream, but robust quantitative OOD scoring is still incomplete because matched references are unavailable for many prompts; therefore the main reported metrics remain in-distribution. The absolute accuracy gap from real data and discriminator AUC above 0.5 suggest opportunities for improvement via larger DiT architectures, hard-negative mining in CLOP training, or multi-resolution conditioning schemes. Finally, validation on additional tissues and organisms beyond the current 80-dataset corpus would strengthen generalization claims.

5. Conclusions

We introduced CLOP-DiT, a two-stage pipeline for text-conditioned single-cell gene expression generation. By combining contrastive language–omics pretraining (CLOP, 3.02M parameters) with a conditional diffusion transformer (DiT1D, 22.10M parameters) trained via flow matching, CLOP-DiT generates biologically meaningful cell profiles from natural-language descriptions. Across 100 cell types from 80 datasets (220,304 cells), the model achieves 37$\times$ random KNN accuracy, 81% steering accuracy, and a diversity ratio of 0.93 with the recommended Midpoint solver configuration. Downstream biological analyses show partial transfer: clustering/mixing agreement is substantial, classifier transfer reaches 30.8% accuracy (macro F1 0.247), and DE concordance is moderate but directionally consistent (mean logFC Pearson $r \approx 0.39$, mean sign agreement ≈ 0.94 across tested contrasts). CLOP-DiT therefore supports a viable—but not yet fully indistinguishable—text-conditioned generation paradigm for *in silico* single-cell biology, with applications in data augmentation, rare cell-type simulation, and hypothesis-driven perturbation studies.

Supplementary Materials: The following supporting information can be downloaded at: https://www.mdpi.com/article/10.3390/biology1010000/s1. Table S1: GEO accession identifiers for all 80 training datasets; Table S2: held-out validation dataset identifiers; Table S3: full CFG sweep results (8 guidance scales $\times$ 2 solvers $\times$ 3 metrics); Figure S1: per-type KNN accuracy ranked by training-set cell count. Reproduction instructions (environment setup, `regenerate_report.sh`, expected outputs) are provided in `docs/QUICK_START.md` and `scripts/regenerate_report.sh`.

Author Contributions: Z.F.: conceptualization, methodology, software, validation, formal analysis, investigation, data curation, writing—original draft preparation, writing—review and editing, visualization. The author has read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Ethical review and approval were waived for this study, as only publicly available, de-identified data from the Gene Expression Omnibus (GEO) were analyzed. No new human subjects research, animal experiments, or clinical interventions were conducted.

Informed Consent Statement: Not applicable.

Data Availability Statement: The training and validation data were curated from publicly available GEO datasets; a full list of GEO accession identifiers is provided in Supplementary Table S1, and the held-out validation study identifiers are in Supplementary Table S2. The train/validation split configuration is defined in `configs/clop_v9.3.yaml`. Model checkpoints, generated embeddings, and the figure-reproduction script (`scripts/regenerate_report.sh`) are available from the corresponding author upon reasonable request. All evaluation code is available in the project repository (available upon request pending institutional approval for open-source release). Reproduction instructions (one-command figure regeneration) are in `docs/QUICK_START.md`.

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Abbreviations

The following abbreviations are used in this manuscript:

CLOP	Contrastive Language-Omics Pretraining
DiT	Diffusion Transformer
scRNA-seq	Single-cell RNA sequencing
CFG	Classifier-free guidance
KNN	K-nearest neighbor
DE	Differential expression
scGPT	Single-cell Generative Pre-trained Transformer

Appendix A. Architecture Details

Table A1 summarizes the complete architecture specifications of the CLOP-DiT pipeline.

Table A1. Architecture specifications.

Component	Architecture	Parameters
BiomedBERT-large (frozen)	Transformer encoder	340M
CLOP text projector	MLP [1024→1024→1024→256]	1.58M
CLOP cell projector	MLP [512→512→512→256]	1.44M
DiT1D	8× AdaLN-Zero blocks, 6 heads	22.10M
scGPT encoder (frozen)	Transformer	∼51M
scGPT decoder (frozen)	Gene token prediction head	included

Appendix B. CFG Scale Sweep

Table A2 reports the full CFG sweep across 8 guidance scales with 10-step Euler integration and 50 evaluation groups.

Table A2. CFG scale sweep (10-step Euler, 50 eval groups).

CFG	KNN-1	Steering	DivR	KNN/Random
0.5	0.147	0.788	1.13	15×
1.0	0.348	0.796	0.76	35×
1.5	0.396	0.798	0.60	40×
2.0	0.407	0.806	0.53	41×
2.5	0.408	0.804	0.49	41×
3.0	0.410	0.802	0.48	41×
4.0	0.402	0.804	0.50	40×
5.0	0.380	0.798	0.55	38×

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